

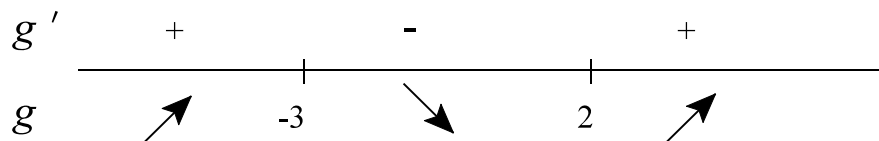
$$g(x) = 4x^3 + 6x^2 - 72x + 9$$

a. Domain = $\mathbb{R}$

b. Intercepts (0,9), (-5.1092,0), (0.1264,0), (3.4828,0)

c. $g'(x) = 12x^2 + 12x - 72 \Rightarrow$ C.N. $x = -3, 2$

d.



g is decreasing on $(-3, 2)$

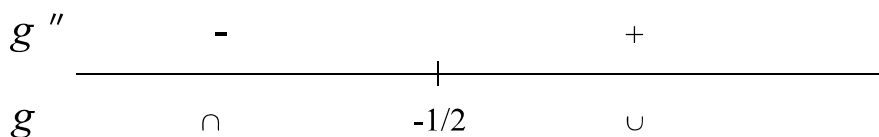
g is increasing on $(-\infty, -3) \cup (2, \infty)$

e. $P(-3, 171)$ is a relative maximum on the graph

$P(2, -79)$ is a relative minimum on the graph

f. $g''(x) = 24x + 12 \Rightarrow$ 2nd C.N. $x = -1/2$

g.



g is concave up on $(-1/2, \infty)$

g is concave down on $(-\infty, -1/2)$

h. Inflection point $P(-1/2, 46)$

i.

-4, 137.

-3, 171.

-2, 145.

-1, 83.

0, 9.

1, -53.

2, -79.

3, -45.

4, 73.

j.

